TABLE II

VARIANCE DECOMPOSITION AND REALLOCATION EXERCISE IN THE STATIC MODEL[a]

Variance Decomposition (×100)				
$\frac{\mathrm{Var}(\alpha)}{\mathrm{Var}(y)}$	$\frac{\mathrm{Var}(\psi)}{\mathrm{Var}(y)}$	$\frac{2\,\mathrm{Cov}(\alpha,\psi)}{\mathrm{Var}(y)}$	$\frac{\mathrm{Var}(\varepsilon)}{\mathrm{Var}(y)}$	$\mathrm{Corr}(\alpha,\psi)$
60.03	2.56	12.17	25.24	49.13
(0.85)	(0.16)	(0.39)	(0.59)	(0.86)
Reallocation Exercise (×100)				
Mean	Median	10%-Quantile	90%-Quantile	Variance
0.50	0.58	2.60	−1.24	−1.12
(0.10)	(0.11)	(0.19)	(0.31)	(0.11)

[a]Estimates of the static model, on 2002–2004. In the top panel, α denotes the worker effect, and ψ denotes the firm effect, in the linear regression $Y = \alpha + \psi + \varepsilon$. In the bottom panel, we report differences in means, quantiles, and variances of log-earnings between two samples: a counterfactual sample where workers are randomly reallocated to firms, and the original sample. The results are obtained using 1,000,000 simulations, and we report parametric bootstrap standard errors in parentheses (computed using 200 replications).

presence of some complementarity between firms and lower-type workers, those complementarities explain only a small part of the variance of log-earnings.

We next consider the impact on log-earnings of a reallocation exercise where workers are randomly allocated to firms. With this exercise, our aim is to assess the contribution of sorting to the distribution of earnings. We assume that earnings functions, for all worker types and firm classes, are not affected by the reallocation, thus abstracting from equilibrium effects. We show the results of the reallocation in the bottom panel of Table II. In the first column, we report the estimate of the difference in mean log-earnings between two samples: a counterfactual sample where workers are randomly allocated among firms, and our original sample. In an additively separable economy between workers and firms, such as under the AKM model, there should be no effect of the reallocation on mean log-earnings (e.g., Graham, Imbens, and Ridder (2014)). However, using our estimates that account for the presence of complementarities, we find a positive mean impact (0.5%), which suggests that the effect of complementarities on average log-earnings is statistically significant but quantitatively small. The positive sign is in line with the fact that the mean worker type tends to increase with firm class k, while complementarities are somewhat stronger in low-wage firm classes.[12]

Moreover, in our distributional framework, we are able to estimate the entire counterfactual earnings distribution corresponding to a given reallocation of workers to firms. In columns 2 to 4 of the bottom panel of Table II, we show the differences in medians and 10% and 90% percentiles of log-earnings between the counterfactual random allocation and our sample. We also report differences in variances in the last column. We see that, while the median effect is in line with the mean effect, the bottom of the distribution would tend to benefit in the random allocation, whereas the top would be hurt. Those differences reflect both the fact that log-earnings are less dispersed in the random alloca-

[12]To provide an intuition, consider the regression model (1). In this specification, the difference between mean outcomes in a population where workers are randomly allocated to firms and in our data is, abstracting from time indices for clarity, $\mathbb{E}^{\mathrm{random}}(Y_i) - \mathbb{E}(Y_i) = -\mathrm{Cov}(b(k_i), \mathbb{E}(\alpha_i|k_i))$.

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tion, as shown by the reduction in variance, as well as the presence of complementarities at the bottom of the distribution.

Interpretation

The coexistence of strong sorting and weak complementarities may be surprising in the perspective of the search matching models inspired by Becker (1973). In such models, assortative matching patterns arise from complementarities in production, which are often reflected in non-monotonic wages. For example, this happens in the model of Shimer and Smith (2000) with on-the-job search that we have simulated. More generally, the results in this section are difficult to reconcile with models based on revealed preferences for wages only. This may indicate that workers or firms care about other job attributes (Hwang, Mortensen, and Reed (1998)), or that workers of a similar type enjoy working together as in the peer effects literature. Workers and firms might also be more complex than assumed in most models of sorting (Lindenlaub (2017), Lise and Postel-Vinay (2015)). More specifically to the Swedish context, as we discuss in Appendix C, institutions such as unions may contribute to explain why different firms do not pay similar workers very differently.

Robustness Analysis

In order to explore the robustness of our results, we perform several exercises that we summarize in Table III. In panels B and C of the table, we show the results of the variance decomposition for different values of the numbers of firm classes and worker types, respectively. The results are quite stable across K values. In addition, while taking $L = 3$ seems to understate the contribution of worker heterogeneity and overstate that of firm heterogeneity, the results are stable between $L = 5$ and $L = 9$.

In panel D, we report results corresponding to several additional specifications. In the first row, we show the results for a model where log-earnings are distributed as three-component mixtures of Gaussians, conditional on worker types and firm classes. The results are very similar to the baseline. In the second row, we report the results of a variance decomposition applied to residuals of log-earnings on a third-order polynomial in age, three education categories, three sectors (manufacturing, retail, and services), and time indicators, with a set of interactions. The R^2 in the regression is 21%. We estimate the parameters on the regression residuals using our two-step approach, including the k-means step. We find that the contribution of worker unobserved heterogeneity is 50% of the variance, compared to 60% in the baseline. Moreover, the relative contribution of firm heterogeneity is somewhat larger than in the baseline. In rows three and four, we report decompositions estimated on smaller firms (less than 50 workers per firm in 2002) and large firms (more than 50). In smaller firms, the contribution of worker heterogeneity is reduced compared to that of firm heterogeneity and the covariance term, while in larger firms, worker heterogeneity accounts for a greater share of the log-earnings variance. In the fifth row, we show that the variance decomposition implied by a fully nonstationary model gives very close results to our baseline estimates.

In panel E of Table III, we show results based on the interactive regression model (1).[13] This model is a valuable complement to our baseline model for several reasons. First, in the regression model, we do *not* assume that worker types are discrete. Second, estimation only relies on mean and covariance restrictions, not on other features of the

[13]We do not account for covariates (i.e., $c_t = 0$), and we keep $b_t(k)$ constant over time.

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TABLE III

VARIANCE DECOMPOSITION ($\times 100$), STATIC MODEL, ROBUSTNESS CHECKS[a]

	$\frac{\mathrm{Var}(\alpha)}{\mathrm{Var}(y)}$	$\frac{\mathrm{Var}(\psi)}{\mathrm{Var}(y)}$	$\frac{2\,\mathrm{Cov}(\alpha,\psi)}{\mathrm{Var}(y)}$	$\frac{\mathrm{Var}(\varepsilon)}{\mathrm{Var}(y)}$	$\mathrm{Corr}(\alpha,\psi)$
			A. Baseline Model		
	60.0	2.6	12.2	25.2	49.1
			B. Varying the Number of Firm Classes		
$K=3$	65.9	1.4	8.7	24.1	45.3
$K=20$	57.6	2.9	12.2	27.3	47.4
			C. Varying the Number of Worker Types		
$L=3$	13.5	13.4	12.2	60.8	45.4
$L=5$	55.5	3.2	12.8	28.5	47.9
$L=9$	53.3	3.5	13.1	30.1	48.2
			D. Other Mixture Specifications		
Mixture-of-Mixtures	62.6	2.4	11.3	23.8	46.4
Log-Earnings Residuals	49.6	3.5	10.6	36.3	40.0
Firms With $\leq$50 Workers	52.6	4.0	17.0	26.4	59.1
Firms With $>$ 50 Workers	60.1	2.6	7.9	29.4	31.6
Fully Nonstationary	60.5	2.8	12.5	24.3	48.3
			E. Regression Models		
Interactive	56.5	2.1	10.8	30.6	50.2
Linear	60.6	1.7	10.0	27.6	48.5
			F. Other Classifications		
Classify Using Means	57.1	4.2	13.7	25.0	43.8
Split by Percent of Movers	67.2	1.5	8.5	22.9	42.5
Split by Value Added	55.4	3.4	12.5	28.7	46.0
			G. Reclassification (Starting at Baseline)		
1 Iteration	55.9	4.1	13.3	26.7	43.9
10 Iterations	56.3	4.1	12.5	27.2	41.2

[a]Estimates of the static model, on 2002–2004. α denotes the worker effect, and ψ denotes the firm effect, in the linear regression $Y = \alpha + \psi + \varepsilon$. The results are obtained using 1,000,000 simulations. The various specifications are described in the text.

distribution of the data. For example, the parameters $a_t(k)$ and $b(k)$ are estimated from mean restrictions alone, which do not rely on assumptions on serial dependence. In addition, the regression estimator is straightforward to compute. On the other hand, unlike our baseline specification, the model's functional forms restrict the shape of interaction effects between worker and firm heterogeneity. In the second row, we report the results for a regression model where we additionally impose that $b(k) = 1$ for all k. The results in panel E show similar variance decompositions as in our baseline model.

In panel F, we consider several specifications in which we vary the way we classify firms. In the first row, we use deciles of mean log-earnings for the classification. In the next two rows, we split each class in two equally-sized subclasses according to the percentage of job movers in the firm (second row), and value added (third row). These specifications capture other information, beyond differences in log-earnings distributions, contained in mobility patterns and value added. The results show some differences with our baseline estimates. For example, when splitting the classes according to value added, the firm effects variance becomes 3.4%, compared to 2.6% in the baseline. The three specifications in panel F also give slightly smaller correlations between worker and firm effects. However, the differences compared to our baseline estimates are small.

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Last, in panel G of Table III, we show the results of a model-based reclassification. Unlike our baseline classification of firms, this method incorporates information from both periods, including earnings associated with job mobility. We report the result of one and ten iterations, starting from the baseline parameter values. The iterated estimates tend to imply a somewhat larger firm effect and smaller correlation coefficient than our baseline two-step results, although the differences are quantitatively small. We have also performed a similar reclassification starting from different initial classifications: deciles of the firm effects estimates of Abowd, Kramarz, and Margolis (1999), deciles of log value added per worker, deciles of the poaching rank measure of Bagger and Lentz (2014), and deciles of the firm-specific shares of job movers. We have found that, while the initial values of the variance components differ substantially, the values after ten iterations are quite close to each other (see Table S7 in the Supplemental Material).

The magnitudes we find differ from the fixed-effects estimates of Abowd, Kramarz, and Margolis (1999). In particular, the AKM estimate of the variance of firm effects is 32% on our sample, and the estimated correlation coefficient is negative. In short panels with low mobility rates, it is well-documented that the AKM estimator may be biased (Andrews et al. (2008)). Our experiments on simulated data, which we described in Section 4, suggest that biases may be substantial in our case. In additional robustness checks, we have found that the large AKM estimate of the firm effects variance is mostly driven by firms with very small number of job movers, and that the variance drops substantially when using existing bias-correction methods (see Table S2 in the Supplemental Material). These various exercises support our finding that, once netted out of incidental parameter bias, the variance of firm effects is small in the Swedish data.

We have performed two additional robustness checks. To further address the robustness of our classification of firms, we have estimated a two-sided random-effects specification where both worker types and firm classes are treated as stochastic. Compared to our two-step approach, this specification is more challenging to estimate. To check sensitivity to departures from discrete firm heterogeneity, we have computed estimates allowing for within-class firm heterogeneity. In both cases, we have found magnitudes in line with our baseline estimates. We provide details on these robustness checks in the Supplemental Material.

In all our specifications, we consistently find magnitudes that are in line with our baseline estimates. In particular, firm effects net of worker composition explain less than 5% of the log-earnings variance, and the correlation between worker and firm effects ranges between 30% and 50%. Compared to estimates based on AKM—not corrected for bias—in other countries, our findings on Swedish data suggest a smaller contribution of firms to wage dispersion and a stronger association between worker and firm heterogeneity. We are aware of only a handful of structural estimates in the literature. Notably, Bagger and Lentz (2014) found a variance of firm effects of 11% and a correlation coefficient of 32% on Danish data, and Hagedorn, Law, and Manovskii (2017) estimated a correlation between worker and firm types of 75% on a German sample.

6. EMPIRICAL RESULTS II: DYNAMIC MODEL

In this section, we present empirical results for our dynamic model. We start by describing parameter estimates, and we then assess their implications in terms of dynamic patterns.

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